

4.1 Random variables

Def A real-valued function $X: S \rightarrow R$ is called a random variable of the experiment if, for each interval $I \subseteq R$, $\{s : X(s) \in I\}$ is an event.

In probability, the set $\{s : X(s) \in I\}$ is often abbreviated as $\{X \in I\}$ or simply as $X \in I$

- Ex** Three balls are selected randomly without replacement from a box containing 20 balls number 1 through 20. What is the probability that the largest number of the drawn balls is as large as or larger than 17?

Sol: Let X denote the largest drawn number

Sol. (a)

$$P(X = k) = \frac{\binom{k-1}{2}}{\binom{20}{3}}, k = 3, 4, \dots, 20 \Rightarrow P(X \geq 17) = \sum_{k=17}^{20} P(X = k)$$

Sol. (b)

$$P(X \geq 17) = 1 - P(X \leq 16) = 1 - \frac{\binom{16}{3}}{\binom{20}{3}},$$

4.2 Distribution functions

CDF: (**cumulative**) distribution function

$$F_x(t) = F(t) = P(X \leq t)$$

1. F is nondecreasing; that is, if $t < u$, then $F(t) \leq F(u)$.

$$2. \quad \lim_{t \rightarrow \infty} F(t) = 1 \quad \lim_{t \rightarrow -\infty} F(t) = 0$$

3. F is right continuous; that is, for every t in $\mathbb{R}$, $F(t^+) = F(t)$

Event concerning X	Probability in terms of F	Event concerning X	Probability in terms of F
$X \leq a$	$F(a)$	$a < X \leq b$	$F(b) - F(a)$
$X > a$	$1 - F(a)$	$a < X < b$	$F(b^-) - F(a)$
$X < a$	$F(a^-)$	$a \leq X \leq b$	$F(b) - F(a^-)$
$X \geq a$	$1 - F(a^-)$	$a \leq X < b$	$F(b^-) - F(a^-)$
$X = a$	$F(a) - F(a^-)$		

4.3 Discrete random variables

Discrete (離散的)

A set of values that is countable is called discrete

Countable: 有空隙在每個數值之間, 才可以數

Examples of set measure

- Finite: $\{0, 1, 2\}$
- Finite: $\{1, 2, 3, 4, \dots\}$
- Infinite: $\{x : x > 0\}$ (You can not find what is the next of zero, 0.1 0.01 ?)

P.F.M (Probability mass function)

(a) $p(x) = 0$ if $x \notin \{x_1, x_2, x_3, \dots\}$

(b) $p(x_i) = P(X = x_i)$ and hence $p(x_i) \geq 0$

(c) $\sum_{i=1}^{\infty} p(x_i) = 1$

4.4 Expectations of discrete random variables

Expected value

$$E(X) = \sum_{x \in A} xp(x)$$

Called: mean, expectation of X

Denoted: EX , μ_x , μ

Ex 4.19 The tanks of a country's army are numbered 1 to N. In a war this country loses n random tanks to the enemy, who discovers that the captured tanks are numbered. If $X_1, X_2, \dots, X_n$ are the numbers of the captured tanks, what is $E(\max X_i)$? How can the enemy use $E(\max X_i)$ to find an estimate of N, the total number of this country's tanks?

Let $Y = \max X_i$ for $k \in n, n+1, n+2, \dots, N$

$$P(Y = k) = \frac{C_{n-1}^{k-1}}{C_n^N}$$

$$E(Y) = \sum_{k=n}^N k \times P(Y = k)$$

$$= \sum_{k=n}^N k \times \frac{(k-1)!}{(n-1)! \cdot (k-n)!} \times \frac{1}{C_n^N}$$

$$= n \cdot \frac{1}{C_n^N} \cdot \sum_{k=n}^N \frac{k!}{n! \cdot (k-n)!}$$

$$= n \cdot \frac{1}{C_n^N} \cdot \sum_{k=n}^N C_n^k$$

$$= n \cdot \frac{1}{C_n^N} \cdot (C_n^n + C_n^{n+1} + C_n^{n+2} + \dots + C_n^N)$$

$$= n \cdot \frac{1}{C_n^N} \cdot C_{n+1}^{N+1} \neq$$

$$N = \frac{1}{n} \cdot C_n^N \cdot \frac{1}{C_{n+1}^{N+1}} \cdot E(Y) \quad \leftarrow \begin{array}{l} E(Y) \text{ 代入觀察到的最大編號} \\ n \text{ 代入樣本數量} \end{array}$$

$$= \frac{n+1}{n} E(Y) - 1$$

$$\begin{aligned} &= C_n^n + C_n^{n+1} + \dots + C_n^N \\ &= C_{n+1}^{n+1} + C_n^{n+1} \\ &= C_{n+1}^{n+2} + C_n^{n+2} \\ &= C_n^{n+3} + C_n^{n+3} \\ &\vdots \\ &= C_{n+1}^{N+1} \end{aligned}$$

Theorem 4.1

If X is a random variable and $P(X=c)=1$, then $E(X) = c$

Theorem 4.2

$$E[g(x)] = \sum_{x \in A} g(x)p(x)$$

Coro

$E(X)$ obeys the linear rules: $E(aX + b) = aE(X) + b$

4.5 Variances and moments of discrete random variables

Definition: Variance of X

$$\text{Var}(X) = \sigma_x^2 = E[(X - E(X))^2] = E[(E - \mu)^2] = \sum_x (x - \mu)^2 p(x)$$

Standard deviation of X: $\sigma_x = \sqrt{\text{Var}(x)}$

Theorem 4.3

$$\text{Var}(X) = E(X^2) - (E(X))^2$$

Application: $(E(X))^2 \leq E(X^2)$

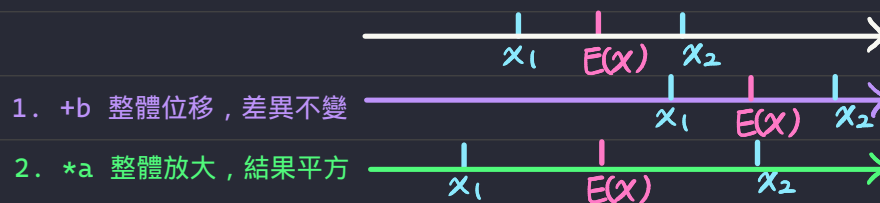
Theorem 4.4

$\text{Var}(X)=0$ if and only if X is constant with probability 1

Theorem 4.5

$$\text{Var}(aX + b) = a^2 \text{Var}(X)$$

→ Var 在計算實際情與平均值的差異（距離）



Definition

Let w be a given point. X is more concentrated about w than is Y . For all $t > 0$

$$P(|Y - w| \leq t) \leq P(|X - w| \leq t)$$

Theorem 4.6

Suppose that $E(X) = E(Y) = a$

If X is more concentrated about a than is Y , then $\text{Var}(X) \leq \text{Var}(Y)$

Definition

$E[g(X)]$	Definition
$E(X^n)$	The n^{th} moment of X
$E(X ^r)$	The r^{th} absolute moment of X
$E(X - c)$	The 1^{st} moment of X about c
$E[(X - c)^n]$	The n^{th} moment of X about c
$E[(X - E(X))^n]$	The n^{th} central moment of X

4.6 Standardized random variables

Definition

The standardized X denote $X^* = (X - \mu)/\sigma$ where μ is the mean σ is 標準差

$$E(X^*) = E\left(\frac{1}{\sigma}X - \frac{\mu}{\sigma}\right) = \frac{1}{\sigma}E(X) - \frac{\mu}{\sigma} = 0$$

$$Var(X^*) = Var\left(\frac{1}{\sigma}X - \frac{\mu}{\sigma}\right) = \frac{1}{\sigma^2}Var(X) = 1$$